SECTION 3 · SUBSECTION 3.6
EXPLAINABILITY

3.6.1 INTRODUCTION.

Explainability is the only stage that calls a language model. It selects a bounded set of files, assembles code context from the vector index, builds a per-file prompt with structural metadata, and requests a short architectural explanation over an OpenAI-compatible endpoint. The one idea the reader must leave with is that selection, not generation, determines quality: only twenty files are explained by default, chosen by risk and import centrality (figure 3.6-1).

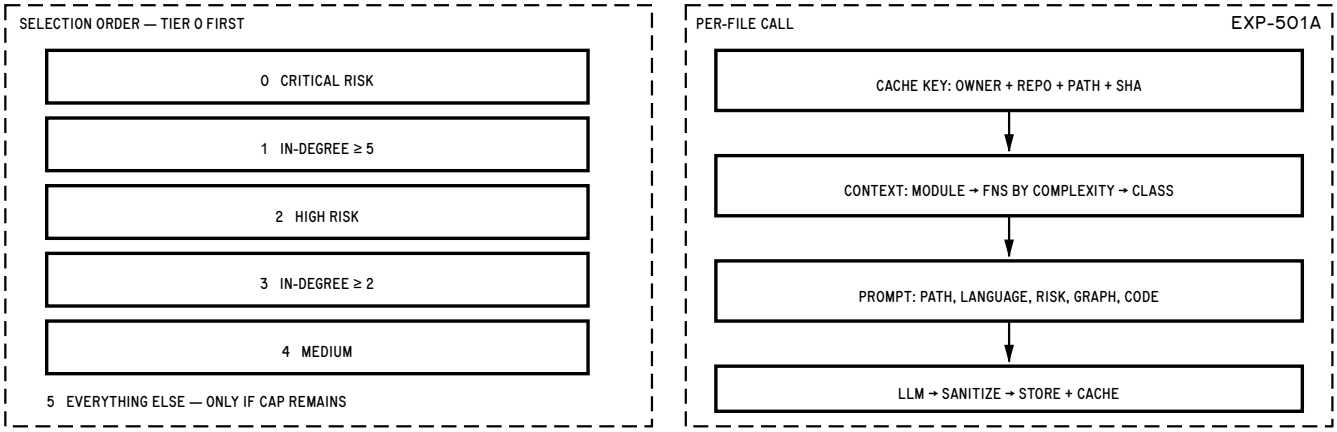


Figure 3.6-1. Selection Tiers and Per-File Call Sequence

3.6.2 FUNCTIONAL DESCRIPTION.

run requires a collection name and an *NVIDIA_API_KEY*; without either it returns the state unchanged. It connects a *CodeRetriever*, reuses explanations for unchanged files in incremental mode, and sorts candidates by tier, then descending in-degree, then path. For each selected file it checks a SHA-keyed disk cache, assembles up to 8,000 characters of chunk text, builds the prompt, and calls the model with temperature 0.1 and an 800-token limit. Successful answers are stripped, stored in state and cached. A failure is counted and skipped; it does not abort the run.

3.6.2.1 Context Budget.

Context assembly orders the module chunk first, then function chunks by descending complexity, then class chunks. It accumulates until 8,000 characters. If fewer than 300 characters remain, it writes a count of omitted chunks instead of a fragment. Retrieved code is therefore always cut at a chunk boundary or marked as truncated, never silently.

```
# src/agents/explainability.py, _assemble_code_context (condensed)
chunks = retriever.get_file_chunks(file_path)          # one file, no semantic query
if not chunks: return raw_content[:8000] if raw_content else ""   # normally empty
ordered = module_chunks + sorted(fn_chunks, key=lambda c: -(c['complexity'] or 0)) + class_chunks
for chunk in ordered:
    if char_count + len(chunk['content']) > 8000:      # hard budget
        parts.append(chunk['content'][:remaining] + "\n... [truncated]")
        break
return "\n\n---\n\n".join(parts)                       # visible separator
```

3.6.3 PERFORMANCE AND DESIGN DATA.

Table 3.6-1. Explanation Parameters

Parameter	Value
Default file cap	20
Allowed cap	0 to 100
Context budget	8,000 characters
High in-degree tier	≥ 5
Medium in-degree tier	≥ 2
Temperature	0.1
Max output tokens	800
Per-call timeout	120 s
Retries / backoff	2 attempts, 1 s base

CAUTION

The stage prints the first eight characters of the API key to its log at start-up. Treat logs as sensitive and remove that line before sharing output. The prompt also contains retrieved repository code; do not send private code to a hosted endpoint without authorisation.

EXPLAIN CHECK	OPERATOR
1 CONFIRM - NVIDIA_API_KEY set	
2 READ - files selected count	
√ CACHE - hit rate reported	
3 OPEN - one explanation	
4 CONFIRM - names real symbols	

EXP